Probabilistic analysis exercises

05-E01 Expectations add despite dependence

Roll two independent fair dice, and let X count sixes. Use indicators to compute E[X] and P(X=1), identifying where independence is needed. How do both values change if the second die always copies the first?

05-E02 Fixed points of a permutation

Uniformly place n named badges into n positions, $n \geq 2$. Find expected fixed points F and the probability that two specified positions are both fixed. Are fixed-point events independent? Handle n=0,1.

05-E03 Six reverse Fisher–Yates paths

On 0-based [a,b,c], for i=2 down to 1 choose j uniformly in [0,i] and swap. List the six (j_2,j_1) paths and outputs. Prove joint uniformity for general n, rather than only uniform positions of individual elements.

05-E04 Expected messages from random splits

A size-n call with $n \geq 2$ prints n stars, chooses uniform p in 1..n−1, and recurses on p,n−p. Size 1 prints an exclamation mark. Derive closed forms for leaves and expected stars, including the adjacent-equation subtraction. Check n=5. Do leaf counts vary between executions?

05-E05 Exactly two hires

Distinct ranks arrive in uniform order; the first person is always hired. Find probabilities of exactly one, two, and n hires. Prove the two-hire formula by disjoint cases. Evaluate at n=6.

05-E06 Marginals and random-path counts

On four elements, choose a cyclic shift uniformly. Prove each element has position probability 1/4, yet the permutation is not uniform. On three elements, prove by path counts that swapping each position with a uniformly selected position from all three is biased. Contrast its invariant with the correct shuffle.

Probabilistic analysis exercises (continued)

05-E07 Bias from priority ties

Assign two named elements independent uniform priorities in {1,2,3,4}; sort stably on ties. Find both output probabilities. Redraw the entire pair on a tie. Prove uniform output and find expected rounds.

05-E08 Nonuniform birthdays

People independently choose date r with probability p_r, summing to 1. Derive one-pair collision probability and expected colliding pairs among k people. Prove uniform dates minimize the pair probability. With four probabilities 1/2,1/4,1/8,1/8, find expected colliding pairs among three people and the exact probability of any collision.

05-E09 Four balls, three bins, and collection

Throw four balls independently and uniformly into three bins. Find expected empty bins, expected singleton bins, and the probability all bins are nonempty. Separately find expected throws from empty bins until all are occupied. Distinguish fixed four throws from random stopping time.

05-E10 Overlapping windows are dependent

Flip eight independent fair coins. Let X count all-H windows of length 3 and L be the longest H run. Find E[X]. Use states for current H-tail lengths 0,1,2 to compute exact $P(L \geq 3)$, listing states at steps 0 through 8. Why do the two answers differ?

05-E11 The optimal threshold for seven candidates

Seven distinct candidates arrive uniformly. Use the strategy that rejects k then selects the first subsequent record. Find optimal integer k and exact success probability, with a conditional-probability derivation and adjacent-difference test. Trace ranks [3,1,6,2,4,7,5] with this k. Does it succeed?

05-E12 Preserve the Quicksort model

For six distinct keys and uniform random pivots, find direct-comparison probabilities for z_2,z_5 and z_1,z_6, and expected total comparisons. If all six keys are equal and two-way partition sends equal keys to the same side, does random pivoting retain the same expected bound?

Probabilistic analysis exercises (continued)

05-E13 Six outcomes from fair bits

Using only independent fair bits, construct an exactly uniform output in 1..6, prove the distribution, and find expected bit count. For three bits directly reduced modulo 6 plus 1, list output preimage counts and the bias.

05-E14 Fair bits from an unknown biased coin

Each independent flip of the same coin is H with unknown fixed probability 0<p<1. Construct an exactly fair bit from pairs of results. Give termination probability and expected flips. Is there a uniform O(1) bound as p approaches 0 or 1?

References and attribution

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein. Introduction to Algorithms, 3rd Edition.

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https://github.com/xinhi6716/computer-science-self-study

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